

Tharun Hrithick SR

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Objective

Fourth-year B.Tech student in Information Science and Engineering with hands-on experience in Python, Generative AI, and prompt engineering. Skilled at working closely with large language models, writing clear domain-specific prompts, and evaluating model outputs with strong attention to detail. Comfortable working independently in remote, asynchronous environments, and equally adaptable to in-office settings, and eager to apply research, reasoning, and analytical skills to help improve the performance of AI systems.

Skills

- **Programming:** Python, Java, JavaScript
- **Web Development:** HTML, CSS
- **Database:** MySQL, PostgreSQL, ChromaDB
- **Tools:** Git, n8n, Claude Code, Postman
- **Technical Interests:** Artificial Intelligence, Agentic AI, Prompt Engineering, Machine Learning, Generative AI, RAG
- **Soft Skills:** Problem-solving, Collaboration, Adaptability, Quick Learning

Projects

Early Lung Cancer Detection Web App

(Completed)

Deep Learning | Flask | PyTorch | CT Scans | SQL

- Designed an end-to-end medical image classification system using Convolutional Neural Networks (CNNs) to detect lung cancer from CT scan images.
- Integrated the trained CNN model with a Flask backend and SQL database, achieving 91% validation accuracy while enabling real-time diagnostic predictions.

Nerve – AI Healthcare Assistant

(Completed)

FastAPI | React | TypeScript | PostgreSQL | ChromaDB | Mistral AI | RAG | OCR

- Built a full-stack AI healthcare assistant using Retrieval-Augmented Generation (RAG) to answer medical queries from uploaded reports, PDFs, and images with context-aware responses.
- Implemented OCR-based document processing, semantic search with ChromaDB, JWT/Google authentication, and persistent chat history using FastAPI and PostgreSQL.

Automated Report Generator

(Completed)

FastAPI | Generative AI | Gemini API | SQLAlchemy | SQLite

- Built an intelligent automated reporting tool that parses active Git repositories to auto-generate structured development summaries via the Gemini 1.5 Flash API.

Emotion Detection & Music Recommendation

(Completed)

Machine Learning | Python | Scikit-Learn | Pandas | NLP | Recommendation Systems

- Conceptualized a real-time sentiment analysis engine to process user inputs and match emotional states to tailored, context-specific audio playlists.

Education

B.Tech in Information Science and Engineering

Jain (Deemed-to-be University), Bangalore

(2023 - 2027)

- Currently in 3rd Year | Aggregate CGPA: 7.55

Paavai Vidhyashram School, Namakkal, Tamil Nadu 637014

Higher Secondary Education(12th) PCMB CBSE, 2023 - 66% | Secondary School Education (10th), 2021 - 69%

Internships -

AI Agent Development Intern — MacAppStudio

- Developed Nerve, an AI-powered Healthcare Assistant using FastAPI, React, and Retrieval-Augmented Generation (RAG) to provide context-aware responses from uploaded medical reports, PDFs, and images.
- Implemented OCR-based document processing with Tesseract, semantic retrieval using ChromaDB, JWT/Google authentication, and PostgreSQL-backed conversation management for secure, persistent user interactions.

Artificial Intelligence virtual intern — Infosys Springboard 6.0

- Engineered a dual-module facial emotion recognition and personalized music recommendation pipeline in Python, optimizing SVM and NLP classifiers to reach 88% accuracy.
- Structured data preprocessing pipelines for metadata normalization, improving overall music recommendation relevance metrics.

Backend Development (Python) — Hawky (Startup), Bangalore

- Architected and optimized high-throughput Flask API endpoints, reducing core application response latencies by 15%.
- Collaborated with startup engineers to build scalable database schemas and resolve critical server bottlenecks prior to product launching.

Certifications -

- Generative AI — Microsoft & LinkedIn
- ChatGPT Prompt Engineering — DeepLearning.AI
- Artificial Intelligence — Infosys Springboard
- Introduction to Agent Skills — Anthropic
- Database and SQL for Data Science in Python — Coursera (IBM)